

Primitive ray-finned fishes

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
ORDERS	Polypteriformes, Acipenseriformes, Lepisosteiformes, Amiiformes
FAMILIES	5
SPECIES	49

Among the least advanced of all bony fishes, primitive ray-finned fishes have an unusual skeleton composed partly of bone and partly of cartilage. There are 4 groups (orders): sturgeons and paddlefishes, gars, bichirs, and the bowfin (a single species that is placed in a separate order). Primitive ray-finned fishes are among the largest of all freshwater species, with sturgeons reaching lengths of 26 ft (8 m). Unique to the Northern

Hemisphere, most species live in freshwater, although some are able to live alternately in fresh and saltwater.

Anatomy

In most primitive ray-finned fishes, the skull and some fin supports are made of bone, while the body and tail are supported by a cartilage notochord with some rudimentary vertebrae. The tail is asymmetrical, with the end of the notochord extending into an upper lobe that is longer than the lower one. Gars, some bichirs, and the bowfin have a swim bladder lined with blood vessels that can be used as a lung when levels of oxygen in the water are low.

Sturgeons are large, bottom-feeding fishes with a flattened snout, a protrusible mouth, and sensitive barbels for locating food. Several



PADDFISHES

There are two species of paddlefishes, both found in large river systems with an ample supply of plankton. The American paddlefish (shown here) lives in the Mississippi river system; it feeds at night and spends the daylight hours resting at the bottom of deep pools. The other species lives in the Yangtze River.

Acipenser sturio

European sturgeon

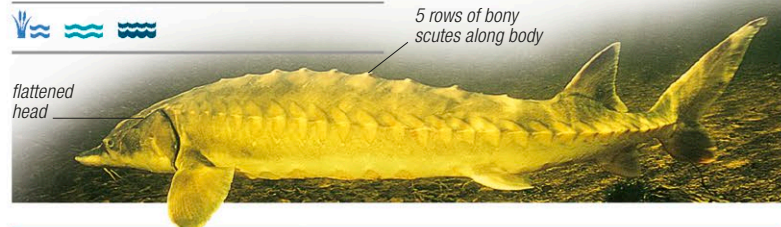


Length Up to 11 ft (3.5 m)
Weight Up to 690 lb (315 kg)
Sex Male/Female
Status Critically endangered

Location Previous known range North Atlantic, Mediterranean, Europe



Heavily overfished for its flesh and eggs—caviar is made of the unshed eggs of females—the European sturgeon is now extremely rare. It used to swim up many European rivers to breed, but spawning in the wild is now restricted to the Garonne River in France. It is greenish brown and, like other sturgeons, has a downward-facing mouth. Estimates suggest fewer than 750 wild fish are left, but this species is now farmed for stocking into rivers.



Polyodon spathula

American paddlefish



Length 4–6 ft (1.2–1.8 m)
Habit Up to 53 lb (24 kg)
Sex Male/Female
Status Vulnerable

Location C. to S.E. North America



One of the few freshwater fishes that feed by straining plankton from water, the American paddlefish is often described as a living plankton net. It sweeps through the water with its lower jaw dropped and the sides of the head inflated to form a funnel-like opening, filtering large amounts of water and extracting plankton with its gills. The underside of its distinctive paddle-shaped snout is covered with electrosensitive pores. This fish has been heavily exploited for its flesh and eggs.



rows of large, thickened scales (bony scutes) protect an otherwise scaleless body. Paddlefishes have a long, paddle-shaped rostrum, which may act as a sensory organ or improve the flow of water to the mouth during feeding. These filter-feeding fishes use their gaping mouth and large gill chambers to strain small aquatic organisms from the water.

Gars are cylindrical and highly predatory fishes that catch prey through ambush or bursts of speed. The elongated rostrum and jaws are

studded with sharp teeth, resembling the snout of a crocodile. Gars are covered by smooth, thickened ganoid scales (see p.468).

Bichirs have a long, slender body protected by thick scales. Most species (sometimes referred to as the true bichirs) have a row of small finlets on their back and a pectoral fin divided into fleshy lobes.

The bowfin is much smaller than other primitive ray-finned fishes. A predatory fish, it has a blunt head and small teeth.

Polypterus ornatipinnus

Ornate bichir

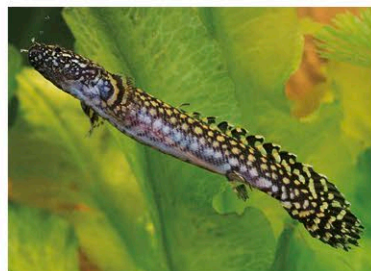


Location W. to C. Africa



Length 16 in (40 cm) or more
Weight Up to 18 oz (500 g)
Sex Male/Female
Status Least concern

Bichirs are cigar-shaped fishes with triangular finlets arranged in a row down the rear half of their backs. They have armorlike scales and tubular nostrils, and can breathe atmospheric air since their swim bladder functions much like a primitive lung. The ornate bichir—one of 14 species in the bichir family—is beige, covered by black, netlike markings. It is a slow swimmer that stalks its prey, such as small fishes, amphibians, and crustaceans, quickly sucking them in when in range.



Lepisosteus osseus

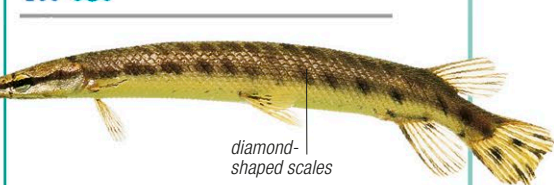
Longnose gar



Location C. to E. North America



Length 4–6 ft (1.2–1.8 m)
Weight Up to 33 lb (15 kg)
Sex Male/Female
Status Least concern



The longnose gar is one of a group of primitive North American fishes that have long jaws with sharp teeth. Its body is long, and its propulsive fins are set far back—much like those of pikes (see p.502). It is primarily a freshwater fish, but in the southern part of its range, adults are frequently found in brackish water. The longnose gar hangs motionless in the water, hidden by vegetation, and waits for prey to come within striking distance. Then, with a sudden thrust, it takes the prey cross-wise in its mouth, often holding it for several minutes. Its predatory habits make it unpopular among fishermen since it can become entangled in nets and cause a good deal of damage.